
METHOD STATEMENT

Pulling of Fire Fighting Pipes on Main Pipe Rack

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1.0 Purpose

This Method Statement defines the safe and controlled procedure for pulling, positioning, and installing firefighting pipes onto the main pipe rack and its branches for the Project. It is issued to the Consultant / Owner's representative for review and approval prior to commencement of the activity and shall be read in conjunction with the applicable project specifications, approved drawings, material submittals, the project HSE Plan and the site-specific Risk Assessment.

2.0 Scope of Work

SIAC will take delivery of pipe materials from the approved supplier, store them in accordance with the manufacturers' and suppliers' recommendations, and install them on the main pipe rack in accordance with the approved schedule of installation methods. The installation shall be inspected for compliance with the relevant project requirements and applicable international standards.

All manpower and materials employed in the installation shall be of the required skill and quality and shall conform to the reference standards; where an item is not fully covered by the contract specifications, the best modern industry practice, as determined and approved by the Engineer, shall apply.

3.0 References / Codes & Standards

The work shall be executed in accordance with, but not limited to, the following references:

- Approved project drawings, piping isometrics and material submittals.
- Project Contract Specifications and Owner's technical requirements.
- Project Health, Safety & Environment (HSE) Plan.
- Manufacturer's / supplier's storage, handling, and installation recommendations.
- Applicable local and international lifting/rigging safety regulations.
- Approved Rigging & Lifting Plan and supporting lift calculations for this activity.

4.0 Definitions & Abbreviations

Term	Definition
MOS	Method of Statement
HSE	Health, Safety & Environment
QHSE	Quality, Health, Safety & Environment
JSA	Job Safety Analysis
PPE	Personal Protective Equipment
Rachit	Manual come along / lever hoist winch used to draw the wire rope and pull the pipe
C.S. Sch40	Carbon Steel, Schedule 40 pipe wall thickness
SWL	Safe Working Load

5.0 Roles & Responsibilities

5.1 Site Engineer

- i. Coordinates the work and explains the task and this method statement to the site supervisors and crew prior to commencement.
- ii. Ensures installation is carried out as per approved specifications, drawings, and manufacturer's recommendations.
- iii. Confirms that materials used are as per the approved material submittal.
- iv. Verifies rigging equipment, roller supports, and chain blocks are inspected, certified and within their safe working load prior to use.

5.2 Safety Officer

- v. Ensures appropriate safety measures are implemented and maintained to avoid any unexpected events during pipe pulling operations.
- vi. Conducts toolbox talk and confirms JSA / permit-to-work requirements are completed before work starts.
- vii. Monitors the exclusion zone beneath and around the pipe rack during pulling and lifting activities.
- viii. Has authority to stop work immediately if an unsafe condition is identified.

5.3 Rigger / Crane Operator

- ix. Holds a valid, current rigging/crane operator certification.
- x. Carries out lifting operations strictly in accordance with the approved Lifting Plan.
- xi. Inspects slings, shackles and lifting gear before each lift and reports any defects.

5.4 Pipe Fitters

- xii. Install and secure the rollers on the pipe rack branch prior to lifting the pipes.
- xiii. Operate the chain blocks and Rachit winch under the direction of the site engineer/supervisor.
- xiv. Maintain constant communication (hand signals / radio) during pulling to ensure controlled, synchronized movement.

6.0 Resources

6.1 Manpower

Designation	Qty	Responsibility
Site Engineer	1	Supervision & technical control
Safety Officer	1	HSE monitoring & control of exclusion zone
Foreman / Supervisor	1	Direct control of crew and sequence of work
Rigger (certified)	2	Rigging, slinging, chain block / Rachit operation
Pipe Fitter	2 – 3	Roller installation, pipe alignment & securing
Crane Operator (certified)	1	Operation of mobile crane
Helper / Labor	2	General assistance, housekeeping, tag lines

Crew numbers may be adjusted by the Site Engineer to suit actual site conditions, provided the minimum supervision and safety coverage stated above is maintained.

6.2 Equipment & Tools

Equipment	Capacity / Spec	Purpose
Chain Block (x2)	3 Ton SWL	Suspending / hanging the pipe at fixed points during pulling
Rachit (Come along / Lever Hoist)	3 Ton SWL	Drawing the wire rope to pull the pipe through the rack
Wire Rope Sling & Wrapping	As per rigging plan	Wrapping/anchoring the pipe to the Rachit
Roller Supports (4 per 12 m span)	580 kg SWL each*	Supporting and guiding the pipe as it is rolled into position
Mobile Crane	As per approved Lifting Plan	Lifting / pulling pipes on main rack
Shackles, Slings, Rigging Hardware	Certified, within SWL	General rigging
Tag Lines	—	Controlling pipe movement / preventing swing
Radios / Hand-signal set	—	Communication between crew during the pull

*Roller design load of 580 kg each is supported by structural calculation with a stress-analysis mechanism failure load of 1,070 Ton each (see Attachment – Roller Design Calculation).

6.3 Materials & PPE

- Utility pipes – Carbon Steel Sch. 40, largest diameter 250 mm (approx. 362 kg per 6 m length), per approved material submittal.
- Standard PPE for all personnel: safety helmet, safety shoes, coverall, high-visibility vest, safety glasses, gloves.
- Task-specific PPE for riggers/personnel working at height: full-body harness with double lanyard, connected to an approved anchor point.

7.0 Methodology / Sequence of Work

7.1 Pre-Work Preparations

1. Obtain approved Permit to Work / Lifting Permit and confirm this MOS, JSA and Lifting Plan have been approved.
2. Conduct toolbox talk with all involved personnel, covering the task sequence, hazards, and controls.
3. Inspect and barricade the work area beneath the pipe rack to establish an exclusion zone, post warning signage.
4. Inspect all rigging equipment (chain blocks, Rachit, slings, shackles) and confirm certification / SWL tags are valid.
5. Install and secure rollers on the pipe rack branch prior to lifting the pipes (4 no. rollers per 12 m span, at the designated fixing points).

This branch has two levels; the largest diameter pipe is 200 mm C.S. Sch. 40 (approx. 280 kg for a 6 m length). Lifting in this direction is performed using two chain blocks of 3-ton capacity together with a 3-ton capacity Rachit.

1. Secure the pipe by hanging it from the 1st (fixed) suspension point on the branch using a chain block.
2. Wrap the pipe from the inside using the wire rope connected to the Rachit (3-ton capacity).
3. Unhook the pipe from the 2nd (external) suspension point near the branch, while keeping the 1st point secured.
4. Begin pulling the pipe from inside the pipe rack in a slow, controlled manner using the Rachit.
5. Once approximately two-thirds (2/3) of the pipe length are inside the pipe rack, release the 2nd external hanging point.
6. Continue rolling the pipe on the installed roller supports up to its final position on the pipe rack.
7. Once in final position, secure the pipe on its permanent supports as per the approved drawings before releasing all rigging.

Notes – Roller System

- Rollers are mounted on the pipe rack branch before the pipes are lifted.
- 4 no. rollers are installed along each 12 m span, at defined fixing points.
- Each roller is designed for a working load of 580 kg (see attached design calculation).
- Stress analysis of the roller mechanism confirms a failure load of 1,070 Ton per roller, providing a substantial safety margin over the working load.

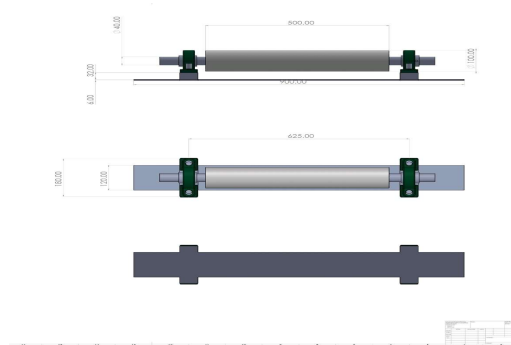


Figure 3 – Roller mechanism / stress analysis reference

8.0 Lifting & Rigging Requirements

- All lifting operations shall be carried out strictly in accordance with the approved project Lifting Plan and calculations.
- All lifting gear (chain blocks, Rachit, slings, shackles) shall be third-party certified, color-coded and within the current inspection validity, with SWL clearly tagged.
- No load shall exceed the rated SWL of the chain blocks (3 Ton), Rachit (3 Ton), or rollers (580 kg each) at any point in the operation.
- A banksman/signalman shall direct all crane movements; only trained and authorized personnel shall operate lifting equipment.
- The exclusion zone beneath the suspended/rolling load shall be always maintained and monitored by the Safety Officer.

9.0 Quality Control & Inspection

- Verify pipe material, size, and schedule against the approved material submittal prior to installation.
- Inspect rollers and supports for correct installation and secure fixing before pulling commences.
- Visually inspect pipe surface/coating after pulling for any damage caused during the operation.
- Confirm final pipe position, alignment and support fixing against the approved drawings; record on the applicable inspection/ITP checklist.
- Any non-conformance shall be recorded and closed out prior to proceeding to the next branch/section.

Document Type Risk Assessment
Project Horus Capacity Expansion Nubaria, Egypt Project, Egypt
Activity SIAC-Pipe rack pipes pulling
Revision 00
Date 29 July 2026

Activity	Potential Hazards	Unmitigated Risk			Recommended Control	Evaluation Risk after Control		
		L	S	R		L	S	R
General	• Manual handling	4	3	12	• Ensure PTW was approved. • Workers are trained in safe handling techniques. • Using safe manual handling technique • Mechanical equipment / manual tools will be used wherever possible. • Ensure the workers not lift more than 25 kilograms/person. • The material handling will be done in the correct method, controlled by trained personnel. • Toolbox Talk to all relevant workers related to manual handling.	1	3	3
	• Ergonomics	4	3	12	• Take Proper Working Postures • Avoid excessive bending. • Shift positions frequently prevent prolonged stress on any single muscle group. • Implement job rotation to reduce repetitive stress injuries.	2	3	6
	• Slipping & tripping	5	3	15	• Good housekeeping and material arrangements to be maintained at all times. • Ensure electrical cables management was done. • Ensure the walkway is clear of obstructions. • Escape routes must keep always clear.	1	3	3
Work at Height	• Falls from height. • Falling objects. • Slips and trips. • Adverse weather conditions.	4	5	20	• Guardrails, personal fall arrest systems (harnesses), and secure edge protection. • Scaffold must be inspected by the scaffolding inspector and green tagged. • The body harness must be inspected and color-coded prior to being used. • TBT to the relevant workers about the hazards and control measures of working at Height. • Tool lanyards, toe boards, debris nets, exclusion zones below, and mandatory hard hats. • Proper housekeeping and material arrangements, anti-slip surfaces, clear walkways, adequate lighting, and safety footwear. • Keep access free from obstacles.	1	5	5

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		L	S	R		L	S	R
					Monitor weather forecasts, suspend work during high winds/rain/lightning, and implement heat/cold stress controls.			
Use of Scaffolds or Working Platforms.	- Collapse or overturning. - Falls from platform. - Falling tools or materials. - overloading or misuse.	4	5	20	- Use only approved scaffolds, erected and inspected by certified scaffolding inspector. - Ensure scaffold tags (green/red) are displayed. - Provide guardrails, mid-rails, and toe boards on all working platforms. - Workers must wear full body harness with double lanyard + shock absorber attached to approved anchorage. - Helmets with chin straps mandatory. - Tools secured with tool lanyards to prevent falling. - Barricade the area below scaffolds with warning signs to protect other workers. - No work at height during high winds, rain, or poor visibility. - Materials/tools lifted by pulley/hoist – not carried while climbing. - Conduct Toolbox Talk daily before work. - Routine inspection of anchorage points, lifelines, and scaffolds. - Maintain good housekeeping on platforms to avoid slips/trips.to climb over the scaffolding ledgers and posts. - Strictly adhere to rated load capacity, distribute loads evenly, prohibit excess material storage on platforms.	1	5	5
Material delivery, unloading & storage	- Collision with persons, - Mechanical failure. - Collision with another vehicle	4	5	20	- Drivers to be competent, trained and own suitable license. - Certified controlled and well-maintained machines. - Making daily inspection checklist. “Documented.” - Making monthly equipment inspection (HSE, Mechanical) “documented” - Limit the number of workers. - Designate a clear walkway far away from equipment movement.	1	5	5

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		L	S	R		L	S	R
					- Vehicles to be fitted with mirrors and additionally to be provided with banksman when reversing. - Equipment speed limit within the work area to 10 km/h - Vehicles fitted with audible reversing alarms - All operatives wear high-visibility clothing. - Check ground condition. - Ensure good visibility. - Windscreen is regularly cleaned up. - Check underground installation presence prior to earthwork. - Wearing Mandatory PPE is obligatory.			
Use of power tools (Drilling Machines & Hammer drills)	• Electricity	4	5	20	- Ensure the inspection was done before use & only used if in safe conditions & color coded. - Ensure the electrical cables elevated and not scattered at the access routes. - The electricity supply will be protected by suitable safety devices. (GFCI) - Electrical supply leads will be in good condition and color coded. - Prohibited to connect the earth clamp with MEWP or scaffold. - Portable Fire Extinguisher to be in close proximity to work area. - All sockets and plugs should be IP65 for construction use. - Powered hand tools are inspected and protected with a grounding system.	1	5	5
	• Inhalation of Dust and Particles • Excessive Noise	5	3	15	- Personnel working in areas with noise in excess of 85dBA to wear hearing protection. - Personnel must wear full personal protective equipment (PPE) in addition to a clear face shield. - Personnel must wear proper type of mask	1	3	3
	• Sharp edges • striking fixed object	4	3	12	- TBT should be provided for the relevant workers. - A proper type of glove shall be provided and worn.	2	3	6

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		L	S	R		L	S	R
					- Only trained/qualified technicians are allowed to perform the activity. - Provide a clear walkway for workers, free from any obstructions or protruding parts. - Provide adequate lighting in the area if required to ensure clear visibility for workers. - Clearly mark sharp edges and hazardous areas using warning signs / tape			
	• Vibration	5	3	15	- Gloves are worn. - Toolbox Talks given to operatives on benefits of massaging and exercising fingers during work breaks. - Only trained/qualified technicians are allowed to perform the activity. - Continuous monitoring and proper supervision to determine the appropriate exposure time to prevent vibration hazards.	1	3	3
	• Contact with rotating parts (entanglement).	5	3	15	Avoid loose clothing, keep hands clear of the chuck/bit.	1	3	3
Use of hand tools: - (Measuring Tape & Spirit Level Torque Wrench, Spanners and Socket Wrench Set, Beam Clamps Installation Tools & Nuts, Bolts, Washers).	- Sharp edges. - Dropped objects hazard. - Hand and finger injuries. - Tool slippage. - Flying debris/shards - Eye injuries.	5	4	20	- Provide proper gloves, ensure all hand tools are inspected & free of defects before use, clear waste/shavings using brushes or magnetic collectors (never bare hands). - Use tool lanyards & tethering systems to secure hand tools, install toe boards or exclusion zones below, secure the area beneath the tie-off/work point, use a dedicated tools box/tool tray to hold nuts, bolts, washers, and keep items secured to prevent falling. - Conduct TBT on hand tool safety, ensure technicians are qualified, experienced & authorized, identify & avoid pinch points, use correct grip & hand placement. - Ensure tools are properly sized (correct socket/wrench fit), maintain a firm & stable stance during operation, inspect tools for wear/defects, apply controlled force without over-torquing. - Install safety guards on tools where applicable, clear waste using brushes or magnetic collectors, keep the work zone clean & organized. - Wear safety glasses during all tasks.	1	4	4

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		L	S	R		L	S	R
Lifting by (Chain blocks /Ratchet Lever Hoist)	- Load falling - Overloading - Chain failure - Struck by load	4	5	20	- Inspect chain block and valid calibration certificate before use. - Ensure lifting team received proper safety briefings and training. - Do not exceed the safe working load (SWL) - Secure and balance the load properly before lifting. - Keep all workers away from suspended loads. - Use approved lifting tools and accessories only. - Assign a competent person to supervise lifting activities. - Maintain clear communication during lifting operations. - Install hard barricades around the work area	1	5	5
Lifting the Pipe by mobile crane	Falling load / dropped load	4	5	20	- The crane must be inspected, approved and certified, by accredited 3rd party. - The operator must be licensed to operate the mobile crane and must be inducted/trained prior to start working on site. - All the lifting slings and accessories must be inspected, approved and certified by accredited 3rd party. - Lifting PTW must be prepared and approved with all the lifting data, prior to starting the lifting operation. - Check load weight including rigging. - Full exclusion zone and barricades. - The lifting zone must be evacuated. - The lifting operation must be under supervision of competent lifting supervisor. - The crane must be inspected prior to commencing the activity, by using inspection checklist.	1	5	5

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		L	S	R		L	S	R
Crane positioning	• Crane overturning / ground failure	4	5	20	• Verify ground bearing capacity. • Fully deploy outriggers. • Use outrigger mats / steel plates. • Level the crane before lifting. • Follow load chart for actual radius. • Approved lifting permit and approved lifting plan shall be available prior to start of lifting activities.	1	5	5
Slinging and rigging	• Sling failure / wrong angle	4	4	16	• Sling angle not exceeding 60°. • Equal length slings only are allowed being used. • Visual inspection before use must be done by the competent rigger. • Only competent riggers are allowed performing rigging activities. • All lifting gears (slings, shackles...etc) must be inspected and certified by accredited 3rd party. • The slings must be placed under supervision from the lifting supervisor.	1	4	4
Pipe Roller Support	• Hand entrapment • Pipe falling • Roller overturning • Falling object • Slip / Trip hazard	4	5	20	• Wear suitable gloves while handling rollers and pipes. • Ensure the pipe is securely positioned on the rollers before moving or lifting. • Inspect the rollers and confirm they are in good condition before use. • Do not exceed the safe working load (SWL) of the rollers. • Keep the work area clean and free from obstructions to prevent slips and trips.	1	5	5
Disposal of waste	• Environmental pollution • (concrete splashes)	3	3	9	• Waste to be transported to approved landfill site. • Ensure scrape will be transported immediately to the dumping area.	1	3	3
Environment	• High noise levels. • The dust.	5	4	20	• Personnel working in areas with noise in excess of 85dBA to wear hearing protection. • Compulsory wearing of PPE. • Checking the condition of equipment before beginning work	1	4	4

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		L	S	R		L	S	R
					- Dust control system to be implemented (Water browser) - Personnel wear safety glasses. - Personnel wear dust masks. - Ensure the equipment was inspected & doesn't have defectives.			
Actions to be taken in case of emergency.					- All the site workers must be trained on the site ERP. - The emergency phone numbers must be well known by all the site workers and posted at different locations on site. - The rescue team must be always ready on call, with the rescue equipment. - All the rescue tools (fire extinguishers, eye wash, stretcher, first aid kit...etc), must be inspected in a daily basis and ready to be used. - All the workers must be reminded about the nearest assembly points and the escape routes, prior to the start of the activity. In case of fire: - Eyewitness tends to extinguish the fire by himself (if he trained for) and informs his direct supervisor about that fact. - The direct supervisor must proceed to the area of fire, assess the situation, and inform the Project & HSE Managers about the fire. - The ERT (Emergency Response Team) shall arrive and apply the ERP rules. In case of medical emergency: - The first person to witness shall warn his supervisor and ask for assistance. - In case the Medical Emergency is due to an accident, check your own safety before approaching the scene of the accident. - Remove the victim to a safer place ONLY in case there is a chance of further injury; else do not touch the victim unless trained to do so. - If you are an approved First Aider, provide first aid, else wait till a First Aider or the site nurse arrives and provides first aid as appropriate. - Continue First Aid treatment till the Male Nurse / Medical Team arrives and takes care			

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		L	S	R		L	S	R
					of the victim. • Assist the Male Nurse / ERT till the Emergency is brought under control. In case of spillage: • If trained, stop flow (if any and investigate the cause of release and inform your supervisor. • The supervisor shall inform the HSE Manager instantly. • Prevent migration of the leak or spill to soil, surface water or ground water using containment methods and/or spill control kits. • Barricade the area of spillage and restrict the entry to the area. • Stop any ignition sources in the vicinity of the release. • Assess the cause and consequence of any environmental damage. • The ERT (Emergency Response Team) shall arrive and apply the ERP rules.			

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Consequence (Severity) Rating Factors (5)					
No	Safety	Health	Environ.	Financial	
				Equip.	Product.
1	First Aid	Immediate	Minor	<SAR 1k	<1 hrs.
2	Medical	Temporary	Short term	<SAR50K	1-2 hrs.
3	Lost Time	Short term	Long term	>SAR300k	2-4 hrs.
4	Disability	Long term	Serious	>SAR500k	4-8 hrs.
5	Fatality	Fatal	Catastrophic	>SAR1m	>1 shift

Probability Rating Factors (5)	
No	Frequency
A (1)	Rare Never occurred
B (2)	Possible Has occurred
C (3)	Likely Has occurred more than once
D (4)	Often Occurs several times per year
E (5)	Frequent/Almost Certain Occurs Frequently

Consequence (Severity)	Frequency/Probability				
	A	B	C	D	E
R = C x P	1	2	3	4	5
5	5	10	15	20	25
4	4	8	12	16	20
3	3	6	9	12	15
2	2	4	6	8	10
1	1	2	3	4	5

SCORE	RESULT	
1-3	Low Risk	No action required unless an escalation of risk is possible.
4-6	Moderate Risk	An activity can operate the subject of management and /or modification.
8-12	High Risk	The activity should be modified to include remedial planning and action and be subject to detailed HSE assessment.
15-25	Extreme Risk	The activity should not proceed in the current form.

Name	Title	Prepared	Checked	Approved	Signatures
Hosam El Deghady	MEP Construction Manager	☒	☐	☐	
Amr MAKRAM	HSE Manager	☐	☒	☐	
Sameh HABIB	Project Manager	☐	☐	☒	